

Nassau Candy Distributor

Factory Reallocation & Shipping Optimization Recommendation System

Objective

Develop a decision-support system that predicts order lead time and evaluates potential factory reassignment scenarios while retaining visibility into historical profitability.

Approach

- Clean and validate the supplied transactional dataset.
- Define Lead Time as Ship Date minus Order Date.
- Engineer calendar and profitability features.
- Compare Linear Regression, Random Forest, and Gradient Boosting using MAE, RMSE, and R^2 .
- Evaluate factory what-if scenarios using the five supplied factories.
- Use historical gross margin as profitability context.
- Provide an interactive Streamlit dashboard.

Model finding

Initial held-out evaluation showed Random Forest performing better than the other required models. The reproducible final comparison is generated by `models/train_model.py`.

Critical limitation

The supplied data contains no historical factory assignments, transportation costs, or factory capacity. The factory module therefore provides transparent scenario estimates rather than causal factory predictions. Absolute logistics cost savings and capacity optimization cannot be calculated from the supplied data.

Business value

For a selected product, region, and ship mode, the dashboard compares the current factory with all five factory scenarios, reports estimated lead-time improvement, provides historical gross-margin context, and flags scenarios with limited historical support.

Future data required

Historical factory assignment, shipment origin, customer coordinates, freight cost, factory capacity, production processing time, and service-level constraints would allow a future version to use a true constrained optimization model.